



Reliability in Few-Shot Classification with Frozen CNN Adapters: A Controlled Study

A controlled study of which design choices govern which aspects of reliability

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Opening

Different aspects of reliability are governed by different design choices

No single design choice makes a few-shot model reliable on every axis.

Reliability aspect	Governed mainly by
Accuracy	shot count + adapter architecture
Calibration	head type + backbone
OOD detection	the uncertainty scoring rule
Calibration repair	a post-hoc refit: it improves calibration but does not fully fix it

Problem

A model can be accurate and still unreliable

Edge devices must recognise **new classes from 1–5 examples** each.

Full fine-tuning updates and stores all of the backbone's millions of weights, which is expensive in parameters and compute.

Parameter-efficient adapters train a tiny fraction instead and keep accuracy, but what happens to the model's **confidence** is rarely measured.

Accuracy ≠ reliability

A model can be accurate while being **poorly calibrated**, or **unable to recognise unfamiliar inputs**.

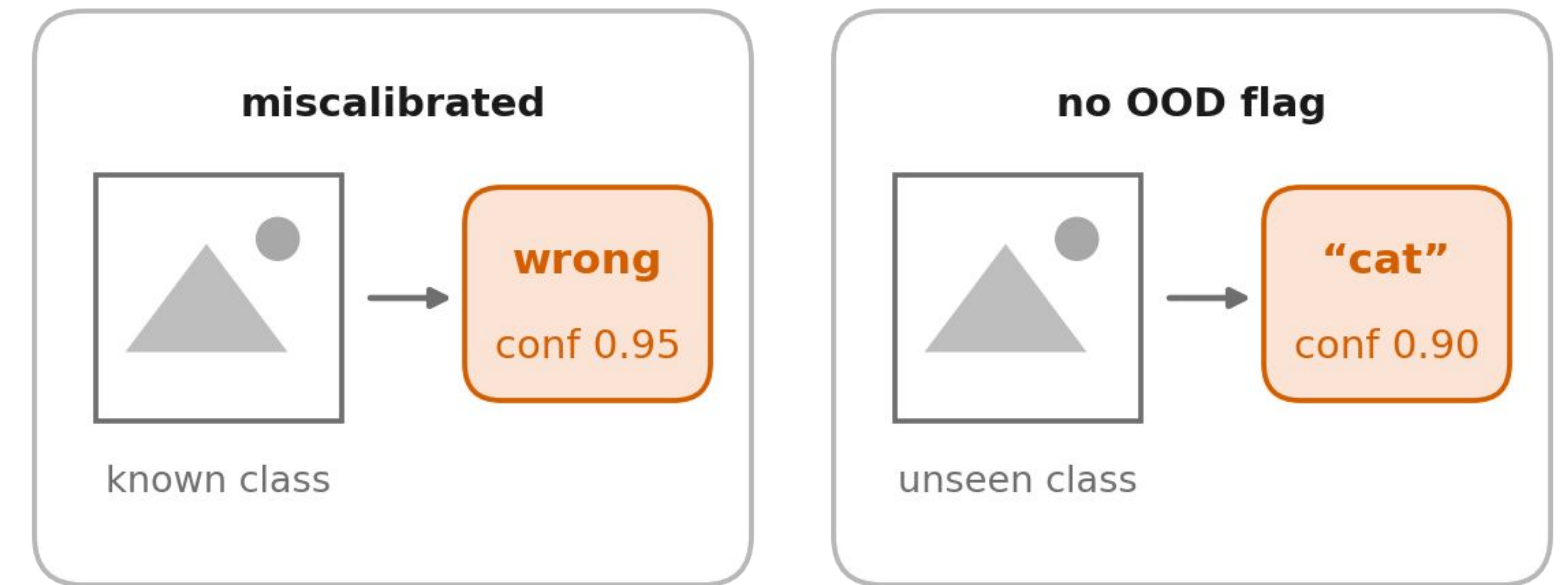


Figure 1 — accuracy vs. reliability (illustration to be added)

Problem statement

Can a tiny trainable adapter make a frozen lightweight CNN reliable for few-shot image classification?

Accurate	Calibrated	OOD-aware
correct on new classes it was not adapted on	its confidence matches how often it is right	flags inputs from outside the task
Challenge	Our response	
Full fine-tuning updates millions of weights	Freeze the backbone; train at most 31.7K adapter parameters once. New classes require no gradient-based adaptation	
PEFT studies rarely measure reliability	Measure accuracy, calibration and OOD detection on the same runs	
PEFT research focuses on transformers	Study adapters inside lightweight CNNs	

Background

Four ideas the talk builds on

1 · FEW-SHOT EPISODE [4]

5 new classes, 1 or 5 labelled images each; classify the rest. Test classes are never used to train the adapter.

3 · EVIDENTIAL (DIRICHLET) HEAD [7]

Outputs **evidence** per class instead of a single probability. Low total evidence $\rightarrow$ high **vacuity** $\rightarrow$ "I don't know".

2 · PARAMETER-EFFICIENT FINE-TUNING [5, 6]

Freeze the pretrained network. Train only a small add-on module: an adapter or LoRA.

4 · MEASURING RELIABILITY

ECE [8]: when the model says 90%, is it right 90% of the time? (lower is better)

OOD AUROC : are unfamiliar inputs scored as more uncertain than familiar ones? (higher is better)

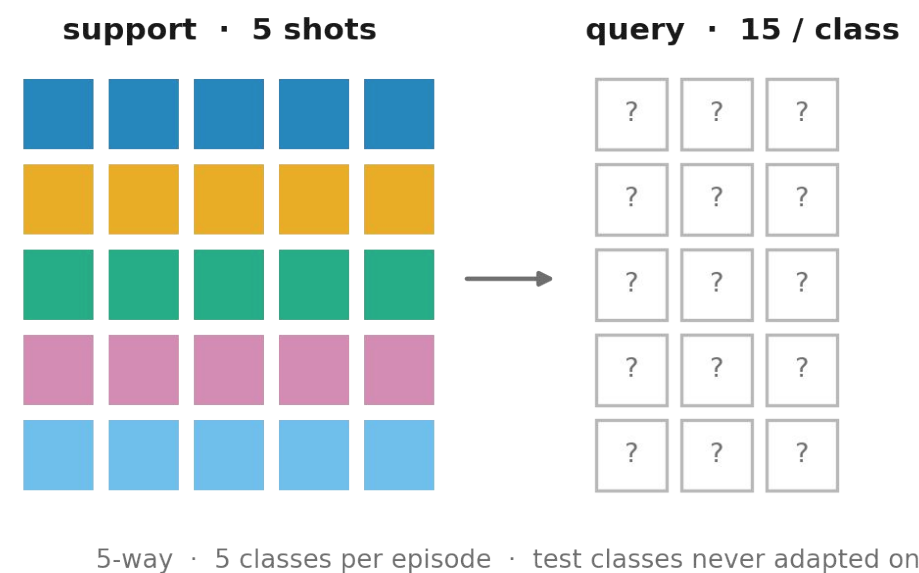


Figure 2 — the few-shot episode: 5 × 5 support grid + query block

Research gap

The gap: properties studied separately, factors changed together

	Frozen CNN backbone	Parameter-efficient adapter	Calibration	OOD detection
TSA [9]	✓	✓	—	—
BEL [10]	— (backbone meta-trained)	—	✓	—
BayesAdapter [11]	— (frozen CLIP ViT)	✓	✓	—
This work	✓	✓	✓	✓

Published comparisons usually change two things at once. We separate them:

training objective vs. OOD score adapter architecture vs. parameter budget

Method

The system: a frozen CNN, a tiny adapter, a parameter-free head

FROZEN BACKBONE

ResNet-18 or MobileNetV3-Small,
never updated

TRAINABLE ADAPTER

6.9K–31.7K parameters, the main
trainable part (+2 scalars in the
evidential head)

PARAMETER-FREE PROTOTYPE HEAD [12]

a class = the average embedding of
its support images

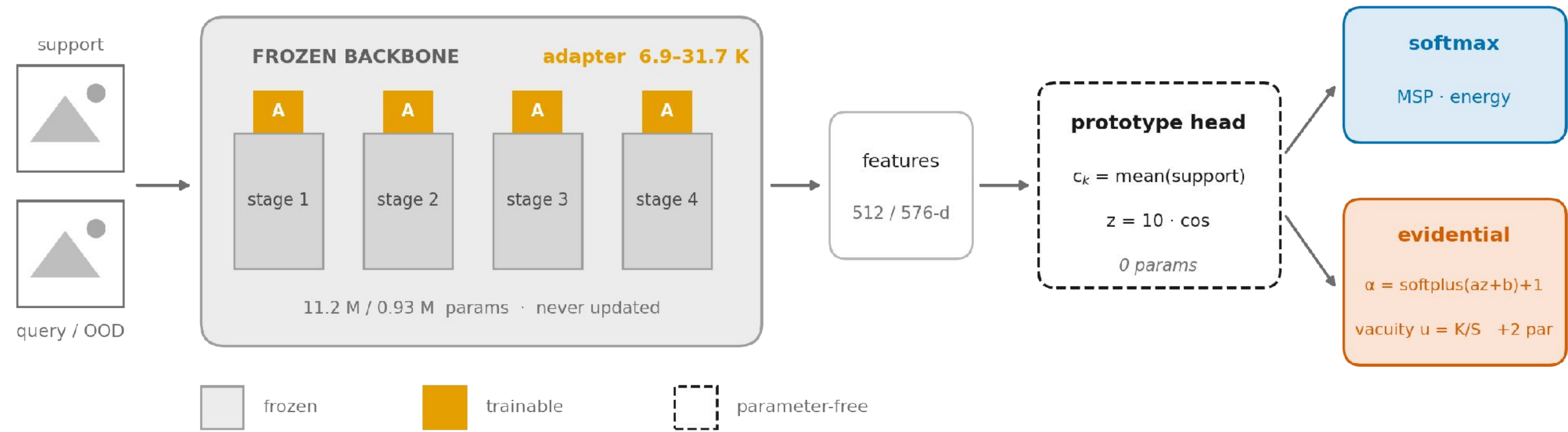


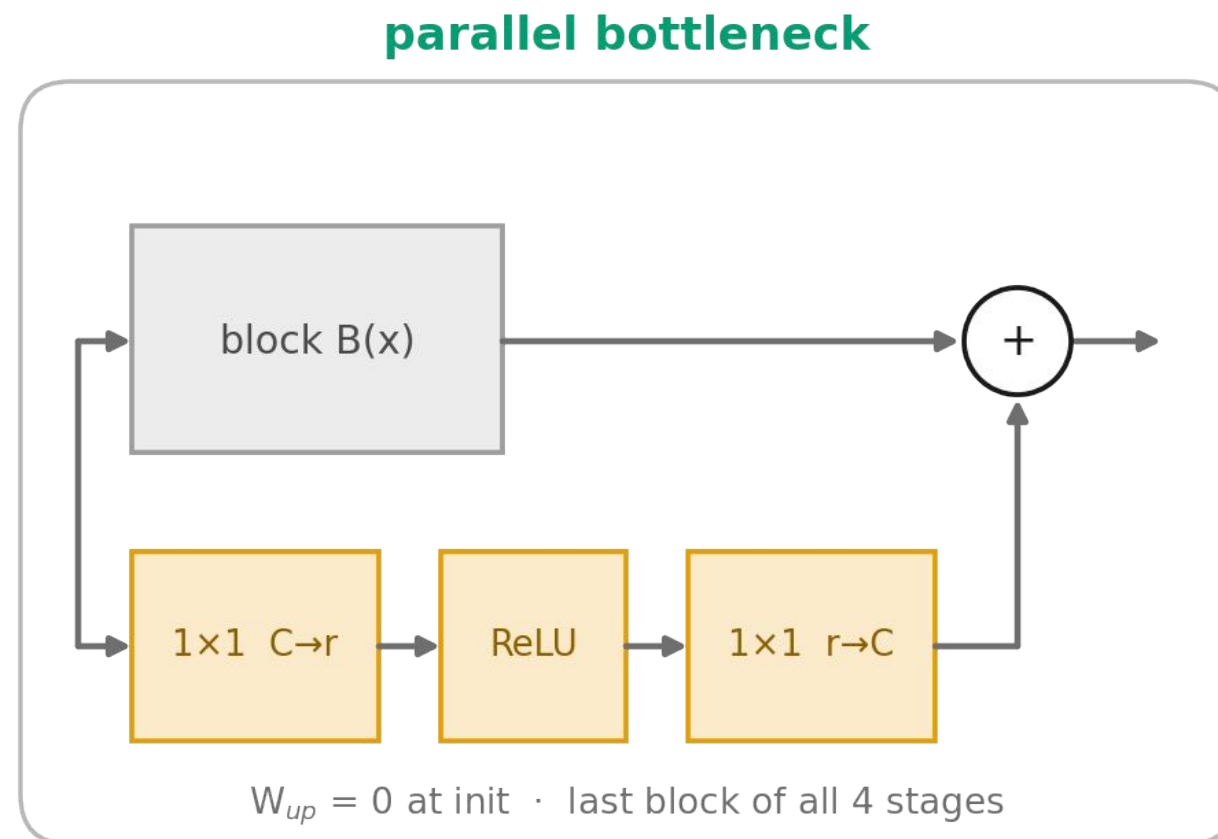
Figure 3 — architecture pipeline (full width)

Method

Two adapter designs: a side branch vs. a low-rank weight update

Parallel bottleneck [5]

A small side branch (down-project $\rightarrow$ ReLU $\rightarrow$ up-project), added at the end of **every stage**



LoRA [6]

A low-rank update **inside one frozen 1 $\times$ 1 convolution** :

$$W = W_0 + (\alpha/r) \cdot BA$$

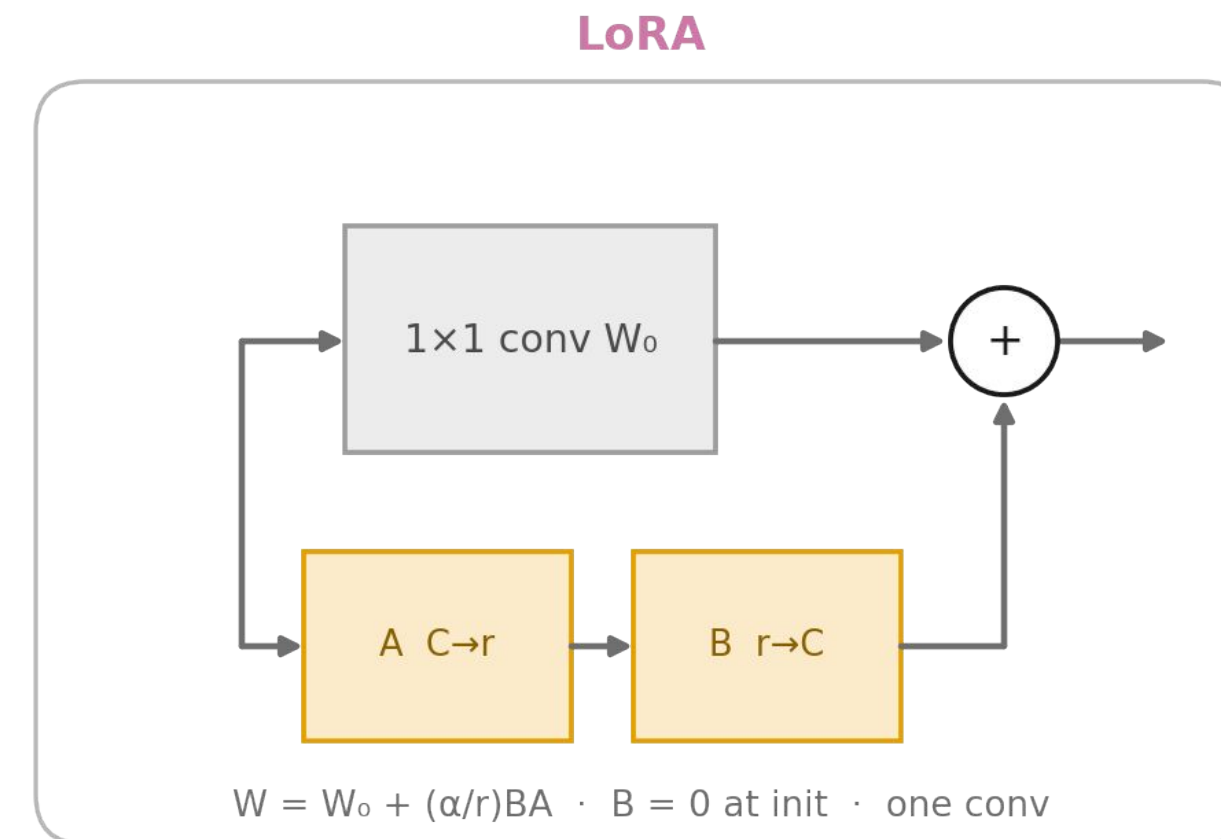


Figure 4 — two adapter zooms, side by side (full width)

Method

One head, two ways to read it

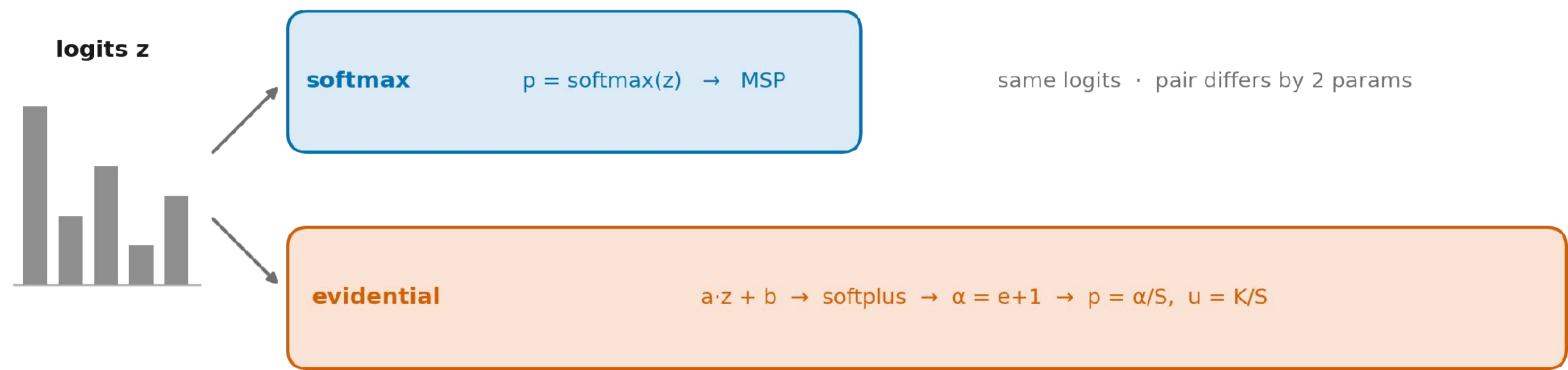


Figure 5 — one set of logits, read two ways

One set of similarity scores (logits), read two ways:

	Softmax	Evidential (Dirichlet)
Output	class probabilities	evidence $\rightarrow$ Dirichlet $\rightarrow$ probabilities
Uncertainty signal	max probability (MSP)	vacuity ("I don't know")
Extra parameters	0	2 (learnable scale and bias)

Experimental design

A controlled factorial design, not one-off comparisons

5 design factors, 2 levels each

Dataset {CIFAR-FS, MiniImageNet} · Shots {1, 5} · Backbone {ResNet-18, MobileNetV3-Small} · Adapter {bottleneck, LoRA} · Head {softmax, evidential}

↓

32 factorial configurations

↓

+ 8 baseline configurations (full fine-tuning, linear probe)

↓

40 configurations × 3 seeds = 120 runs → all 120 models re-scored with 4 OOD scores, plus a post-hoc refit (no retraining)

↓

Matched-budget experiment: 48 runs · **pre-registered**

↓

Rank sweep: 21 runs · looks for a best adapter size (none found in ranks 1–64)

Varying every factor systematically lets us **attribute** differences in each outcome to specific design choices, instead of just ranking models.

Datasets and evaluation

Two few-shot benchmarks, five OOD sets, four reliability measures

IN-DISTRIBUTION

CIFAR-FS [13] · MinilImageNet [14]

5-way, 1-shot or 5-shot

64 / 16 / 20 train / val / test classes (disjoint)

FAR-OOD

SVHN [15] · Gaussian noise

NEAR-OOD

TinyImageNet [16] · held-out CIFAR-100 / MinilImageNet classes

Measure	Question it answers
Accuracy	Is it correct?
ECE	Does its confidence match its correctness?
OOD AUROC	Does it recognise unfamiliar inputs?
Latency · memory · parameters	Can it be deployed?

Research questions

Four research questions

	Question	Answered by
RQ1	Which design choices most strongly explain variation in accuracy, calibration and OOD detection?	120-run factorial grid
RQ2	Is OOD detection explained more by the training objective or by the scoring rule?	All 120 models re-scored with 4 scores
RQ3	Does accuracy follow adapter architecture, and calibration follow parameter budget?	Grid + matched-budget experiment
RQ4	Can evidential calibration be repaired after training without breaking OOD detection?	Post-hoc refit of 60 evidential models

Results · Parameter efficiency

A 31.7K-parameter adapter achieves accuracy comparable to full fine-tuning

TRAINABLE PARAMETERS

Full fine-tuning	11.18M
Bottleneck adapter	31.7K

352× fewer trainable parameters

ACCURACY

Full fine-tuning	90.47%
Bottleneck adapter	91.44%
Linear probe (no adaptation)	87.41%

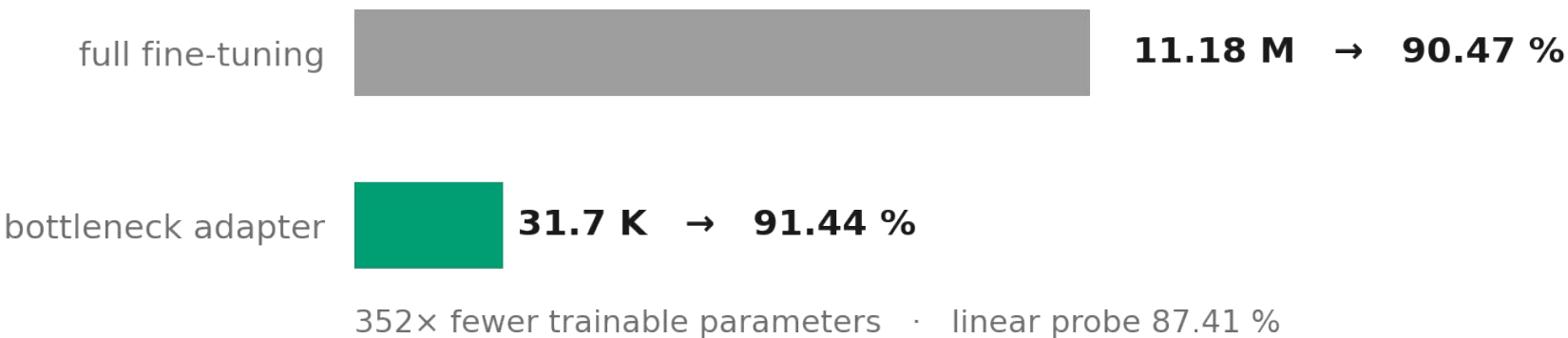


Figure 6 — parameter efficiency (bars)

Results · RQ1

Shots explain accuracy; the head explains calibration

Accuracy → shot count · 76%

Calibration → head type · 83%

OOD detection → no single dominant factor

% = share of the variation across the 32 configurations explained by that factor.

The head changes calibration while barely touching accuracy (0.2% of accuracy variation).

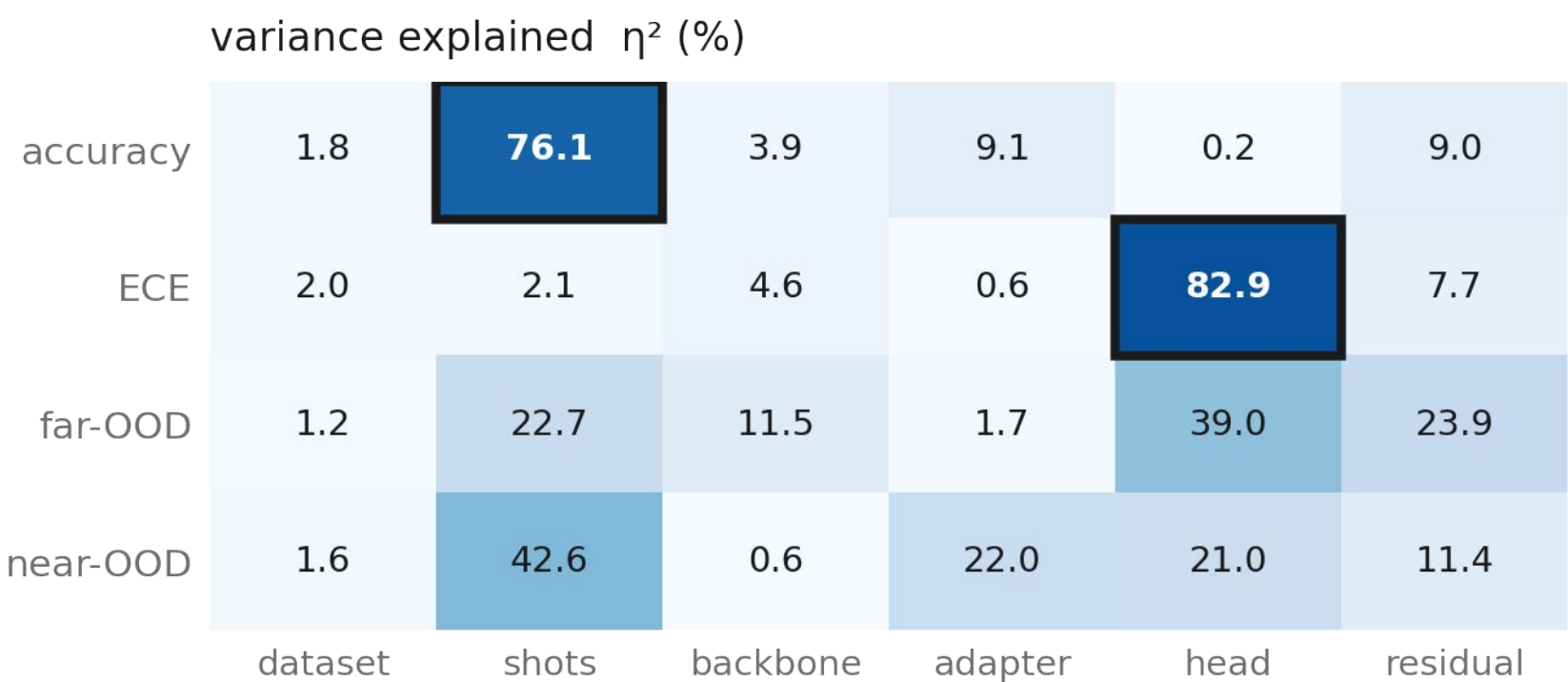


Figure 7 — main-effects heatmap (small)

Results · RQ2

In our setting, OOD detection depends far more on how uncertainty is scored than on how the model was trained

Every model is scored 4 ways, so training objective and score are separated:

MSP [17] · temperature-scaled MSP [8] · energy [18] · vacuity [7]

Far-OOD variation explained: scoring rule 42.7% · training objective 0.17%

Near-OOD: 14.7% vs 0.63%

The same score gives nearly the same AUROC, whichever objective trained the model.

Strongest score: **energy** . It beats vacuity in ~70% of configurations; vacuity beats softmax-probability scores in 37–38 of 40.

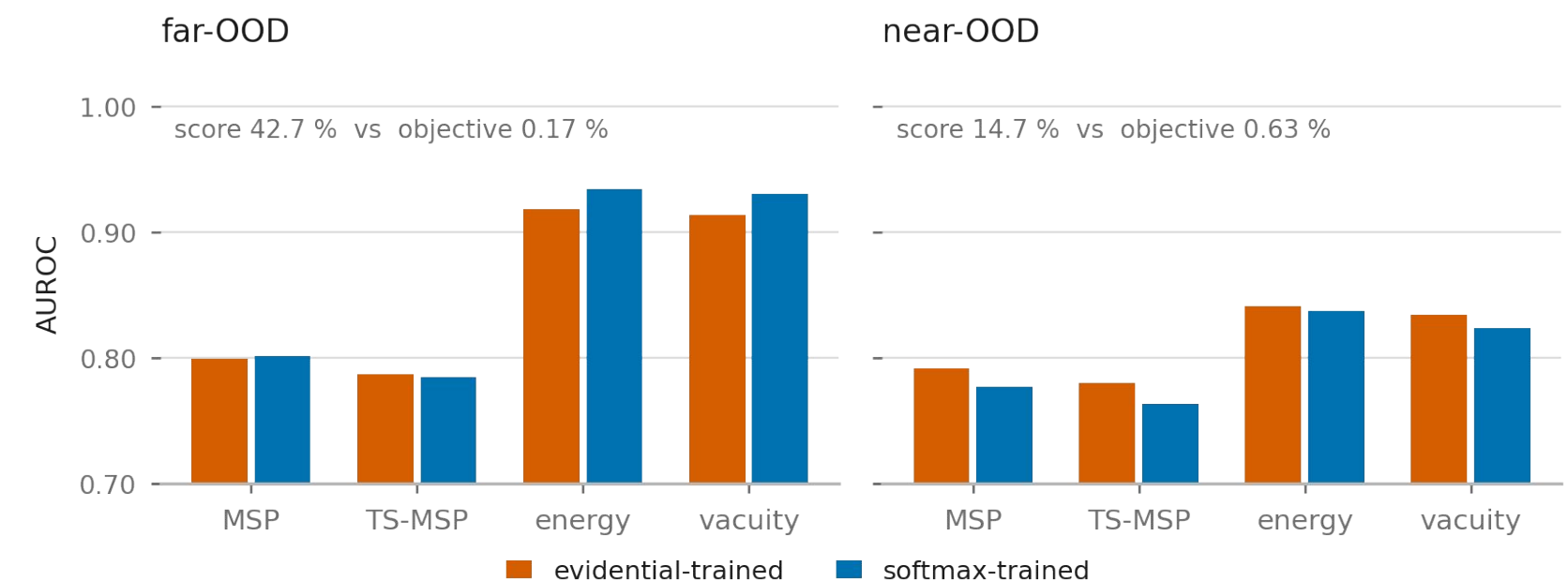


Figure 8 — two panels (far | near)

Results · RQ3

Our first experiment could not separate architecture from budget

Across the 16 matched bottleneck-vs-LoRA pairs in the grid:

Accuracy: bottleneck wins **16/16** , whether it is the larger or the smaller adapter → follows **architecture**

Calibration: the winner **flips between backbones** , and so does which adapter is larger

Backbone	Bottleneck	LoRA	Larger adapter
ResNet-18	31.7K	12.3K	bottleneck
MobileNetV3-Small	6.9K	10.8K	LoRA

Budget and backbone change **together** , so the grid cannot tell which one explains the calibration difference. → **We designed a new experiment.**

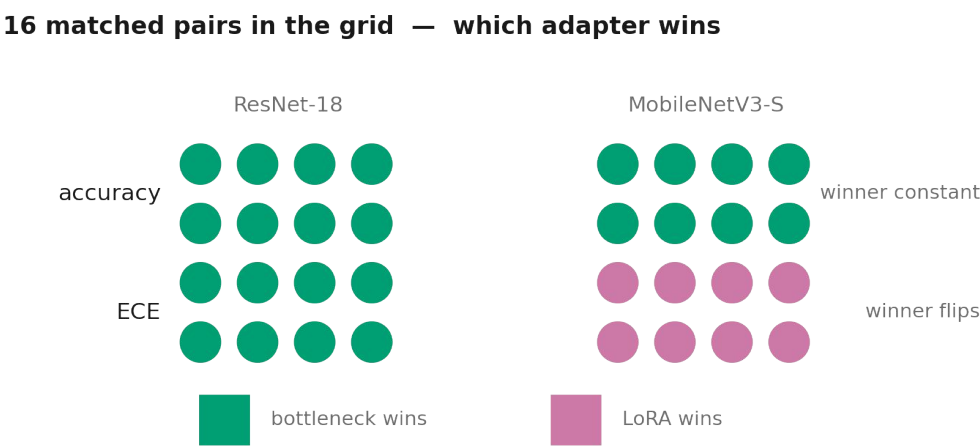


Figure 9 — 16 matched pairs

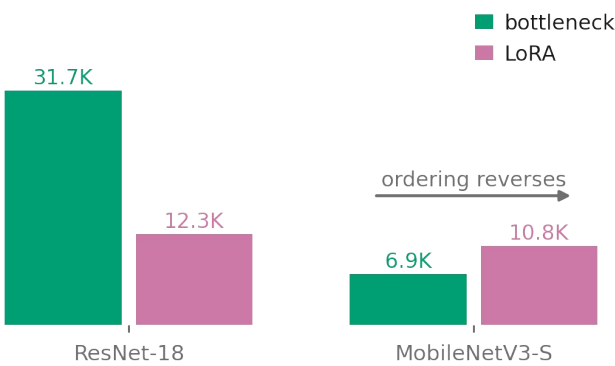


Figure 10 — parameter bars

Results · RQ3 · Matched budget

Matching the budget does not support parameter count as the main explanation

Both adapters rebuilt at the **same budget within each backbone** · MinilmageNet 5-shot · 48 runs · decision rule fixed **before** running

ACCURACY & NEAR-OOD → ARCHITECTURE

Bottleneck still wins 8/8, all beyond 2σ

CALIBRATION → BACKBONE

Budget hypothesis: 0 of 4 cells ·

Backbone dependence: 3 of 4 cells

The calibration gap is unchanged on ResNet-18 and halved (same direction) on MobileNetV3-Small. Which backbone property matters is not yet identified.

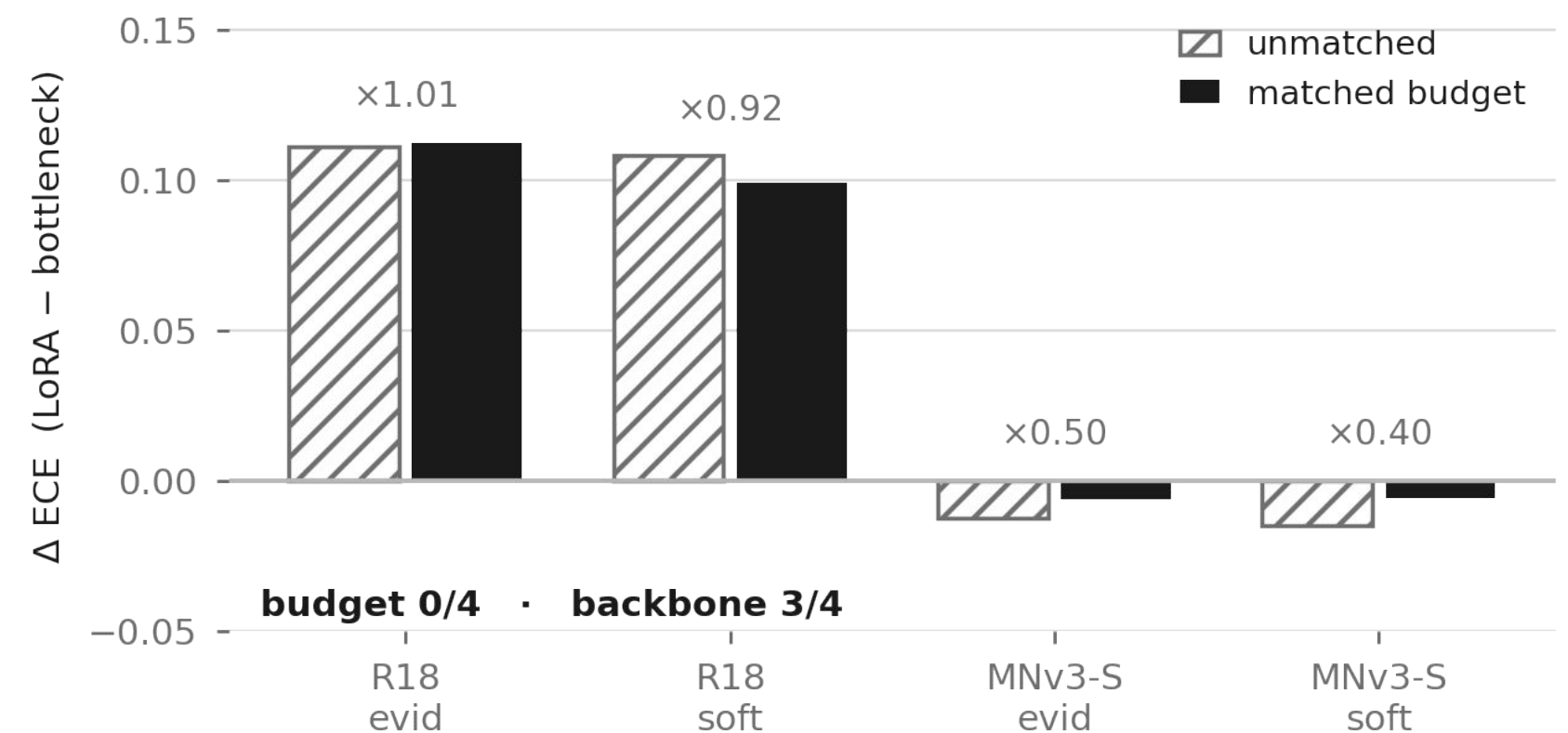


Figure 11 — main visual

Results · RQ4

Refitting two parameters improves evidential calibration, but does not fix it

Refit only the head's 2 evidence parameters on validation episodes · no retraining

ECE 0.323 → 0.185 · improved in 60 / 60 models

Still worse than softmax (0.112) in 60 / 60

OOD ranking preserved in 80% of comparisons (192 / 240)

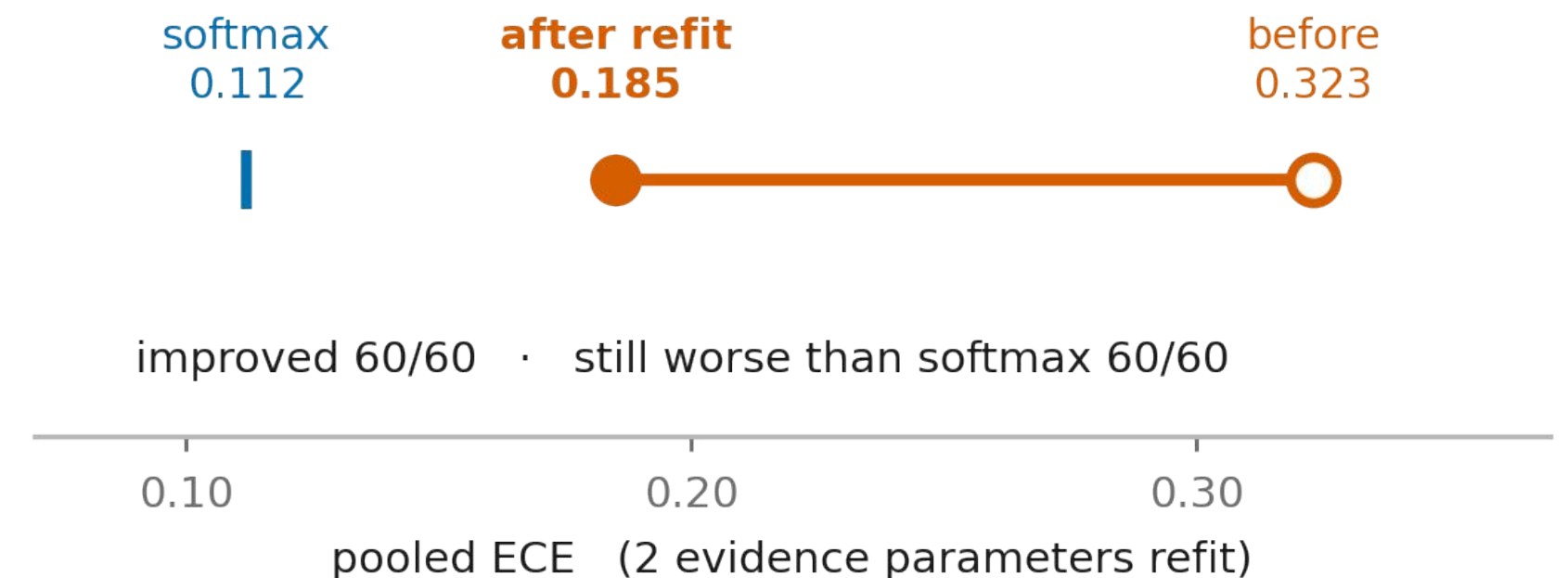


Figure 12 — ECE
before/after

Implications · Deployment cost

Measured inference cost is dominated by the backbone

Design choice	Effect on latency
Backbone (ResNet-18 vs MobileNetV3-Small)	5.1×
Adapter type	4–6%
Evidential vs softmax head	~1%, within measurement noise

In our measurements: backbone dominates latency; adapter choice matters more for accuracy.

Limitations

Limitations and what remains

Pretraining overlap: ImageNet pretraining overlaps MinilImageNet's test classes, which limits comparison with few-shot results from other pretraining setups.

Two backbones: we show that calibration depends on the backbone, not why.

One shared hyperparameter recipe: a controlled comparison, not per-configuration tuning.

Three seeds: small differences should be read cautiously.

MOST IMPORTANT NEXT EXPERIMENT

Add more backbones to identify which backbone property explains the calibration effect.

Contributions

Four separations, and the framework they were run in

Separated adapter architecture from parameter budget with a pre-registered matched-budget experiment

Separated the training objective from the scoring rule for OOD detection, across all 120 models

Decomposed the variation in accuracy, calibration and OOD detection across five design factors

Measured a two-parameter post-hoc recalibration of the evidential head: how much it repairs, and what it costs OOD ranking

Conclusion

What did we learn?

Reliability aspect	Main factor	Evidence
Accuracy	Shots + adapter architecture	76% of variation · bottleneck wins 8/8 at matched budget
Calibration	Head + backbone	83% of variation · backbone dependence 3/4 cells, budget 0/4
OOD detection	Scoring rule	42.7% vs 0.17% of far-OOD variation (score vs objective)
Calibration repair	Post-hoc evidence refit	ECE improved 60/60 · still worse than softmax 60/60

No single design choice optimises every reliability axis.

The main contribution is not a universally superior model, but a controlled decomposition of which design choices govern which aspects of reliability in this few-shot PEFT regime.

Thank you
Questions?

References

Entries [1] – [9]

K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," *CVPR*, 2016.

A. Howard *et al.*, "Searching for MobileNetV3," *ICCV*, 2019.

S. Somvanshi *et al.*, "From Tiny Machine Learning to Tiny Deep Learning: A Survey," *ACM Computing Surveys*, 2025.

O. Vinyals, C. Blundell, T. Lillicrap, K. Kavukcuoglu, and D. Wierstra, "Matching Networks for One Shot Learning," *NIPS*, 2016.

N. Houlsby *et al.*, "Parameter-Efficient Transfer Learning for NLP," *ICML*, 2019.

E. J. Hu *et al.*, "LoRA: Low-Rank Adaptation of Large Language Models," *ICLR*, 2022.

M. Sensoy, L. Kaplan, and M. Kandemir, "Evidential Deep Learning to Quantify Classification Uncertainty," *NIPS*, 2018.

C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On Calibration of Modern Neural Networks," *ICML*, 2017.

W.-H. Li, X. Liu, and H. Bilen, "Cross-domain Few-shot Learning with Task-specific Adapters," *CVPR*, 2022.

References

Entries [10] – [18]

X. Linghu *et al.*, "Bayesian Evidential Learning for Few-Shot Classification," arXiv:2207.13137, 2022.

2. P. Morales-Álvarez, S. Christodoulidis, M. Vakalopoulou, P. Piantanida, and J. Dolz, "BayesAdapter: Enhanced Uncertainty Estimation in CLIP Few-Shot Adaptation," arXiv:2412.09718, 2025.

J. Snell, K. Swersky, and R. S. Zemel, "Prototypical Networks for Few-shot Learning," *NIPS*, 2017.

L. Bertinetto, J. F. Henriques, P. H. S. Torr, and A. Vedaldi, "Meta-learning with Differentiable Closed-form Solvers," *ICLR*, 2019.

S. Ravi and H. Larochelle, "Optimization as a Model for Few-Shot Learning," *ICLR*, 2017.

Y. Netzer *et al.*, "Reading Digits in Natural Images with Unsupervised Feature Learning," *NIPS Workshop on Deep Learning and Unsupervised Feature Learning*, 2011.

7. Y. Le and X. Yang, "Tiny ImageNet Visual Recognition Challenge," Stanford CS231N, 2015.

D. Hendrycks and K. Gimpel, "A Baseline for Detecting Misclassified and Out-of-Distribution Examples in Neural Networks," *ICLR*, 2017.

W. Liu, X. Wang, J. D. Owens, and Y. Li, "Energy-based Out-of-distribution Detection," *NeurIPS*, 2020.